# Homework 2

## 《生物医学图像处理》

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**1** (1) For an image with size $M \times N$, prove that multiplying the image with $(-1)^{x+y}$ and applying Fourier transform, the DC component of the result frequency spectrum will in the center $[\frac{M}{2}, \frac{N}{2}]$.

(2) What will happen to the spectrum if we replace the $(-1)^{x+y}$ to $(-1)^x$

***SOLUTION:***

(1)

$$
\begin{aligned}
F(u,v) &= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(u\frac{x}{M}+v\frac{y}{N})}(-1)^{x+y} \\
&= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(u\frac{x}{M}+v\frac{y}{N})}e^{j\pi x}e^{j\pi y} \\
&= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(u\frac{x}{M}+v\frac{y}{N})}e^{j2\pi(\frac{x}{M}\frac{M}{2}+\frac{y}{N}\frac{N}{2})} \\
&= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi[(u-\frac{M}{2})\frac{x}{M}+(v-\frac{N}{2})\frac{y}{N}]}
\end{aligned}
$$

因此，直流分量在频域平移到中心$[\frac{M}{2}, \frac{N}{2}]$

(2)

$$
\begin{aligned}
F(u,v) &= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(u\frac{x}{M}+v\frac{y}{N})}(-1)^{x} \\
&= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(u\frac{x}{M}+v\frac{y}{N})}e^{j\pi x} \\
&= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi[(u-\frac{M}{2})\frac{x}{M}+v\frac{y}{N}]}
\end{aligned}
$$

因此频域中$u$轴平移$\frac{M}{2}$个单位，中心为$u=\frac{M}{2}$

**2** Consider the images shown.

– The image on the right was obtained by:

(a) computing the DFT and shift DC component to center

(b) inverse the magnitude

(c) computing the inverse DFT

– Explain (mathematically) why the image on the right appears as it does.


***SOLUTION:***

(a)从第一题可得，(a)的操作相当于给$f(x,y) \times (-1)^{x+y}$再做 DFT,

$$f'(x,y) = f(x,y)(-1)^{x+y}$$

$$F'(u,v) = \mathcal{F}\{f'(x,y)\}$$

(b)

$$F''(u,v) = -F'(u,v)$$

(c)

$$f''(x,y) = \mathcal{F}^{-1}\{-F'(u,v)\} = -f(x,y)(-1)^{x+y}$$

在执行 IDFT 时，需要平移回去，因此$f''(x,y) = -f(x,y)$

因此，在最终图像中呈现为黑色变白色，白色变黑色。

**3** (1) Proving that for real image $f(x,y)$, the DFT result $F(u,v)$ satisfies property $F(u,v) = F^*(-u,-v)$

(2) For a 2D MR image, if the $FOV_x = FOV_y = 240mm$, and $\max(k_x) = \max\left(k_y\right) = 120$ $rad/m$,

Calculate the total number of full sampling pixels (total k-space sampling points) required according to Nyquist criterion?

(3) If we use conjugate symmetry property, calculate the minimum number of required k-space sampling pixels for problem 2.

***SOLUTION:***

(1) 对于实数图像 $f(x,y)$，其满足 $f^*(x,y) = f(x,y)$。

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(u\frac{x}{M}+v\frac{y}{N}\right)}$$

我们计算 $F^*(-u,-v)$：

$$\begin{aligned}
F^*(-u,-v) &= \left[\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(-u\frac{x}{M}-v\frac{y}{N}\right)}\right]^* \\
&= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f^*(x,y)e^{j2\pi\left(-u\frac{x}{M}-v\frac{y}{N}\right)} \\
&= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(u\frac{x}{M}+v\frac{y}{N}\right)} \\
&= F(u,v)
\end{aligned}$$

即 $F(u,v) = F^*(-u,-v)$

(2) 根据奈奎斯特采样定理，k 空间的采样间隔需满足：

$$\Delta k_x = \frac{2\pi}{\mathrm{FOV}_x}, \quad \Delta k_y = \frac{2\pi}{\mathrm{FOV}_y}$$

各个方向上所需的完整采样点数 $N$ 为：

$$N_x = \frac{2\max(k_x)}{\Delta k_x} \approx 9.17, \text{向上取整，得}10$$

$$N_y = \frac{2\max\left(k_y\right)}{\Delta k_y} \approx 9.17, \text{向上取整，得}10$$

所以总的全采样点数为：

$$N_{\text{total}} = N_x \times N_y = 100$$

(3) 根据第一问证明的共轭对称性，实数图像在 k 空间是一半冗余的。

假设采样点为-5 到 4，则频域的(0,0),(-5,0),(-5,-5),(0,-5)的共轭对称点为自身。 因此所需点数为

$$N = 4 + \frac{96}{2} = 52$$